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Quantitative evidence of the community of shared future for mankind as a driver of sustainable development in human society

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In an era marked by global economic challenges, an increasing array of non-traditional security threats, and a proliferation of conflicts and issues, China has put forth the building of a community of shared future for mankind as a way to address this question of the times. The theory of the community of shared future for mankind offers a framework to address contemporary issues by promoting equality, mutual benefit, security, justice, openness, innovation, multicultural exchange, and green development. This paper innovatively quantifies the five core principles of the theory of the community of shared future for mankind using the entropy method. It calculates the proxy variable for sustainable development, green total factor productivity, using the super-efficiency Slack-Based Measure model in Data Envelopment Analysis. Baseline regression is conducted on this basis. The Principal Component Analysis method and another Data Envelopment Analysis method, the Malmquist index, are utilized to recalculate the indicators, followed by another round of regression analysis. The results are consistent with the baseline regression. Additionally, Ordinary Least Squares, Partial Least Squares, Maximum Likelihood Estimation, Ridge regression, and Lasso regression are employed to ensure the robustness of the results. The findings indicate a significant positive relationship between the community of a shared future for mankind and the sustainable development of human society, with consistent conclusions drawn from different calculation methods and regression techniques. Through value recognition, international cooperation based on benefit-sharing and mutual reciprocity can guide humanity towards a harmonious coexistence with nature, irrespective of regional and economic disparities. This study provides quantitative evidence of the impact of the community of a shared future for mankind on the sustainable development of human society, filling a gap in existing research. Based on these findings, feasible policy recommendations are proposed to promote sustainable development in human society.

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Introduction

Currently, the unprecedented changes in a century are intertwined with the century's pandemic, leading to emerging security challenges, a struggling global economic recovery, and severe setbacks in worldwide development. Where is the world headed: Peace or war? Progress or regression? Openness or isolation? Cooperation or confrontation? These are the questions that the times have put before us (Xi, 2022). What are the solutions to the challenges of our time? Marx (1995) (1842) once profoundly pointed out that the pressing issues of any age share a common fate with any question founded on solid content. Therefore, the main challenge is not the answer but the question itself. He suggested that questions were the emblem of the times, representing the most practical expression of the zeitgeist. In response to the challenges of this era, China introduced the concept of a community of shared future for mankind (CSFM) in 2012 as a solution to the urgent questions of our time (Hu, 2012). In 2015, at the 70th United Nations General Assembly, the concept of a CSFM was further refined. Its core principles are (1) equitable and mutually beneficial partnerships; (2) a fair and shared security framework; (3) an open, inclusive, and innovative development outlook; (4) diverse cultural exchanges for harmony and mutual learning; and (5) a green and sustainable development model (Xi, 2015). The CSFM embodies the humanistic values of integrity, harmony, and inclusiveness, as well as the principles of social equity and justice. The construction of the CSFM has been incorporated into the resolutions of the United Nations, its Security Council, and the Human Rights Council (China Society for Human Rights Studies, 2018). The CSFM theory emphasizes economic integration, mutual integrity, and value inclusiveness, aligning with the typical value pursuits of most countries. The CSFM theory helps promote a more just, harmonious, and sustainable global development. Thus, the necessity of this paper lies in exploring the practical effectiveness of the CSFM theory in addressing these universal issues. It elucidates how the CSFM theory can effectively address contemporary challenges, particularly in advancing human society's sustainability. This paper is the first to quantify the five core principles of the CSFM. Through quantified indicators, the study demonstrates the potential of the CSFM theory in advancing the sustainable development of human society. To guarantee the broad applicability and universality of the research outcomes, this paper will select 184 countries and regions with varying economic

levels, cultural backgrounds, and political systems as samples from 1991 to 2022. The countries in the sample encompass both developed and developing nations, along with those at various stages of economic growth. By conducting an in-depth analysis of these countries, this study will reveal the applicability and operability of the CSFM theory in different contexts. Through a comprehensive analysis of the practices of these sample countries, this paper will provide a series of policy recommendations. These recommendations will focus on promoting the CSFM theory in different countries and regions to achieve a more just, harmonious, and sustainable global development. Existing research needs a quantitative perspective supported by actual data on the application and effects of this theory at the policy level, which is the innovation of this research. By addressing the gap in the quantitative perspective of the CSFM, subsequent researchers can use the quantified indicators of this research to assess the effects of the CSFM on other vital variables and issues, including economic development and cooperation, social stability, and security. The structure of this study is illustrated in Fig. 1.

Literature review

In human societies, negotiation and consultation are often effective in resolving material interests and political differences. However, a standard set of values must be acknowledged for humanity to acknowledge and embrace each other's cultural ideas. Over more than two thousand years, Western values have evolved to adapt to the historical context of each era, forming part of Western traditional culture. Through his discussions on the ideal state, optimal politics, justice and injustice, citizens, and livelihood, Plato (2003) (380 BC–360 BC) outlined a blueprint for an ideal human society. Aristotle (2018) (Late 4th or early 3rd century BC), by describing the 158 Greek city-states as documented by his students, expounded his famous six-fold classification of political systems. The literature on “post-growth”, “de-growth”, and “post-development” theories, among other social development concepts, has proliferated. A prominent figure in “post-growth” theory, Jackson (2009), argues that the continuous increase in consumption minimizes human happiness and can even hinder it. Human society can only thrive within the ecological limits of our finite planet by fundamentally reducing the impact of economic activities on the environment. The “de-growth” theory, a concept in political economy and social

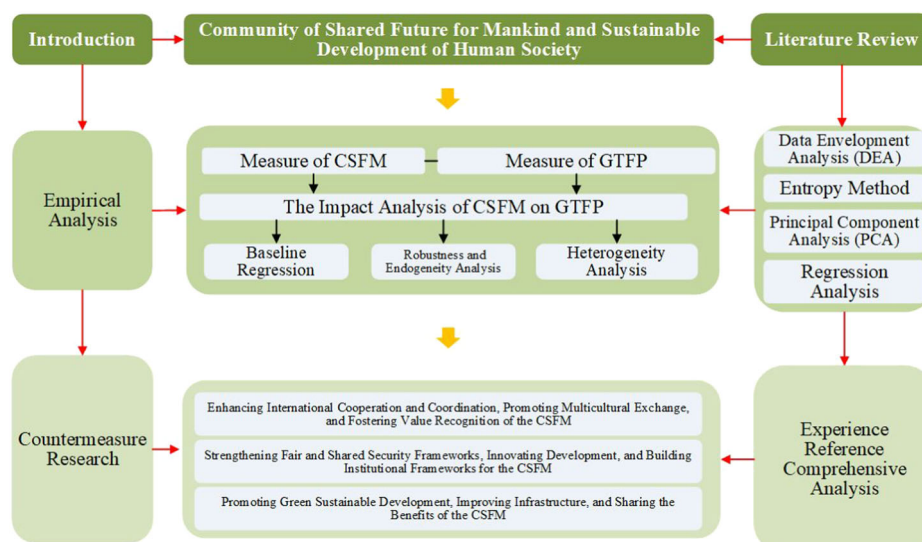


Fig. 1 Research Structure.

movements, is represented by Latouche (2009), who advocates for transforming the existing consumerist model and moving away from dependency on economic growth to build a more sustainable society. “Post-development” is a critical reflection on traditional development theories and practices, questioning the negative effects of economic expansion on the environment, society, and culture. A leading figure in this theory, Escobar (2011), discusses the power relations and cultural politics within development theory and practice from anthropological and post-colonial perspectives. Marx (1958) (1845–1846) asserts that humanity achieves freedom through critical engagement with and transformation of the external world. The realization of self-awareness in freedom can only be achieved within an “association of free individuals”. The CSFM proposed by China is an innovative development of the association of free individuals.

Western scholars have published relatively few articles discussing the concept of the CSFM. Luo & Li (2020) provided a comparative analysis of the study on the CSFM between Chinese and Western academic circles. Li & Shaw (2014) explored the influence of the CSFM on the established international order from the Chinese language and culture perspective. Blackwill & Tellis (2015) argued that the Belt and Road Initiative (BRI) aims to avoid conflict with the United States and the existing international system. Rozman (2014) viewed the ideology of the CSFM as an amalgamation of three elements, with socialism being the foremost factor as a response to the worldwide financial crisis caused by capitalist economies, followed by traditional thought, with Confucianism as one of the traditional sources of the CSFM, starkly contrasting with Western civilization. Huang (2013) saw it as a stance against hegemonism, with the containment strategy adopted by the US and its allies towards China prompting an unprecedented devaluation of the US prospects and the international system by China. Callahan (2016) viewed the establishment of a CSFM as a strategy to reshape the universal order, introducing new governance principles, norms, and rules, while also acting as an instrument to further China’s broader objectives in global governance. Yakunin (2017) pointed out the inclusivity within the concept of the CSFM as key to reshaping globalization. Viehe et al. (2015) viewed this move by China as intended to exclude US influence in the Asia-Pacific region and strengthen control over neighboring countries. Leavy (2018) noted that China’s greatest challenge in garnering support lies in alleviating external concerns over the true nature and extent of China’s leadership ambitions in Asia. Koenig (2019) saw the idea of a CSFM as an important tool for China in leading the world towards more harmonious and peaceful cooperation. Paradise (2016) believed that China, based on its diplomatic policy objectives, embeds values and standards into the CSFM to enhance the influence of its foreign policy, affecting the development of the international system and that of developing countries. Warner (2017) noted that the world economic pattern requires the participation of both the East and West, despite the US’s desire to exit global patterns. Curran (2016); Godbole (2015) considered China’s proposition of the CSFM as a challenge to the current development order. Nie (2016) saw China’s support for overseas investment through the BRI platform as enhancing economic attractiveness, contributing to the construction of a CSFM. The BRI meets China’s developmental needs and those of other participating countries. Murphy (2016) observed that due to the demand for funds by countries along the BRI far exceeding the current supply, China has been adjusting its foreign investment strategy, raising funds from multiple sources to provide soft loans or economic, strategic services for developing financial institutions. Mobley (2019) considered China’s establishment of a series of regional institutions in Asia and Africa with the aim of construct a new order. Hence, the BRI is seen as economic welfare

for countries along its route and as a strategic move by China to seize local resources and expand its influence through diplomatic and economic means. Malik (2017) pointed out that China’s foreign relations goals are usually not to seek direct control but to achieve freedom of action, economic dominance, and diplomatic influence through coercive presence, thus aligning with realist theorists’ expectations of regional hegemony. Liao & McDowell (2015) noted that China has adopted a tougher policy on the South China Sea issue since the end of 2013, but to mitigate the “militarization” controversy, it has sent signals of friendly cooperation to Southeast Asian countries, proposing the BRI as a tool for setting new international rules to stabilize peripheral relations. Suehiro (2017) saw the CSFM as China’s effort to establish a new institutional framework dominated by China, referring to this activity as a “Sinicization process.” Sevilla Jr (2018) considered investment the most effective means for China in constructing the path to a CSFM, using financial influence to shape the behavior of other nations. However, the return on investment raises concerns. Hanauer & Morris (2014) observed that China’s overseas propaganda efforts are influenced by outdated neo-colonialism. Western news agencies, as mouthpieces of Western countries, often smear China’s image with negative reporting to contain its rise. The bias of Western media against China becomes a driving force for China to expand its overseas media influence, aiming to reverse the negative international perception of China’s role. Dollar (2015) saw the CSFM as a challenge to the US’s global dominance and its leadership position in the financial order. Wang (2015) proposed that China needs to seize control of international discourse to mitigate some countries’ negative public opinion on China’s ascent and the internationalization of the concept of a CSFM, reversing the passive situation of China being smeared in international political media.

Only now, a search on the Web of Science for studies focusing on the CSFM and sustainable development yields a relatively small number of 48 research results, as illustrated in Fig. 2. In the current body of research on the sustainable development of the CSFM, analyses have typically focused on individual countries or regions, lacking a global perspective, or have provided only theoretical analysis without empirical data support. For the first time, this paper quantifies the five core principles of CSFM. It employs empirical methods from operations research and econometrics to elucidate the contemporary outcomes of CSFM from a global perspective. Based on this analysis, the paper proposes practical and feasible policy recommendations for the sustainable development of human society.

Method and data

Variable measures

Quantitative indicators for evaluating sustainable development-green total factor productivity. Since the 1960s, climate change, land desertification, resource depletion, and ecological degradation have emerged as critical environmental issues (Chiu & Lee, 2020; Hao et al., 2018; Wang & Lee, 2022; Wen et al., 2022). The development of a green economy has become increasingly prevalent due to climate change and growing environmental threats. Against this backdrop, it is essential to harmonize the economy, resources, and the environment. Naturally, a significant question arises: What indicators are used to assess a country’s performance in mitigating environmental problems, improving environmental performance, and ensuring sustainable economic development? Many economists have conducted in-depth research in this sector. The earliest study was based on the traditional Solow and endogenous growth models. This theory posits that sustainable economies rely solely on technological progress, often measured by total factor productivity (TFP) expansion. However, relevant

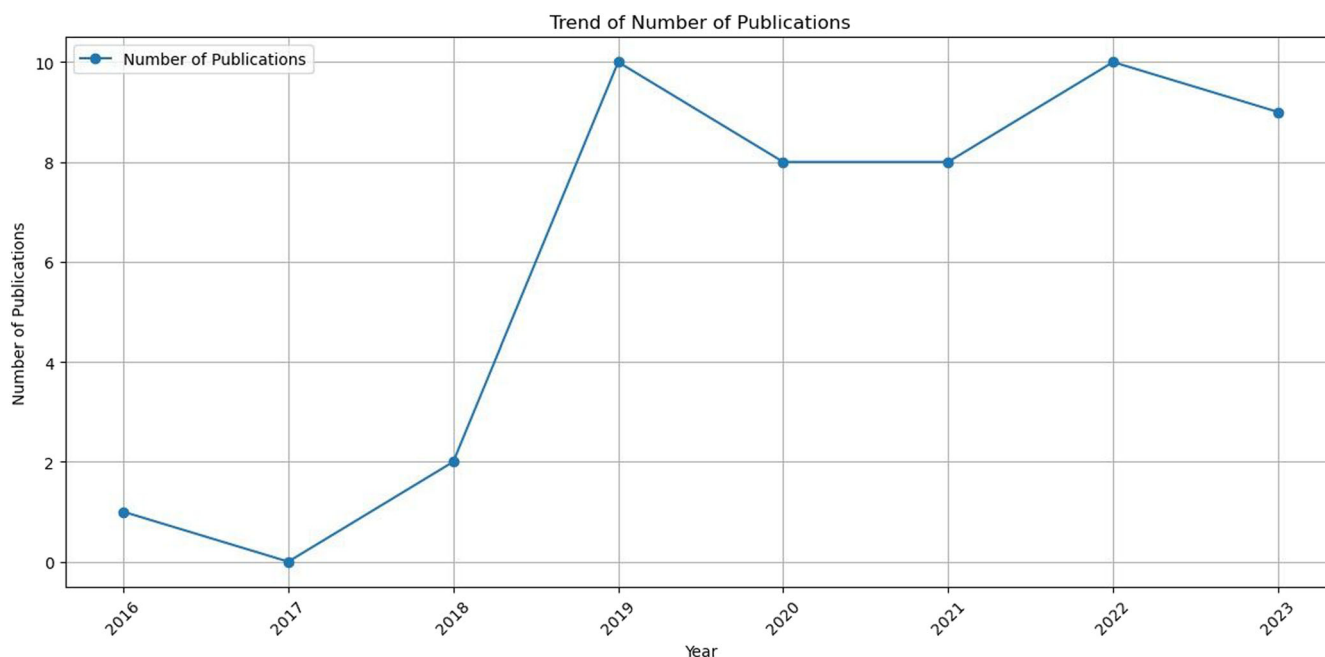


Fig. 2 The Results of the Web of Science Query (Web of Science, 2024).

measurement methods must consider the social benefits of resources, the environment, and other elements (Chen & Golley, 2014; Tian & Lin, 2017). The flaw in this approach becomes apparent whenever energy and the environment become significant constraints on economic growth. Due to the nature of production inputs, some resources are diverted to environmental management activities, resulting in misguided policy recommendations (Liu et al., 2020).

Subsequently, the Asian Productivity Organization introduced the green total factor productivity (GTFP) concept built upon the TFP concept in the 1990s. This introduction attracted considerable academic attention. Since then, researchers have extensively studied GTFP measurement and evaluated whether GTFP growth can assess economic sustainability. For instance, recent studies have considered energy constraints and environmental performance, thereby examining GTFP (Gao et al., 2021; Xia & Xu, 2020; Yan et al., 2020). It is widely recognized that GTFP measures sustainable economic growth more effectively than other economic indicators (Feng et al., 2018; Wang et al., 2022). Therefore, based on previous research, this paper estimates the financial sustainability of 184 countries and regions using the GTFP indicator. In assessing GTFP, it is likewise recognized that limitations related to resources and the environment should be taken into account. For undesirable outputs, most early scholars used undesirable inputs or converted them into negative outputs (Scheel, 2001; Seiford & Zhu, 2002). However, this approach contradicts the production process and can lead to misleading evaluations of green productivity growth. The Directional Distance Function (DDF) method can also be applied. For undesirable outputs, it is assumed that weak disposability applies. Nevertheless, the DDF method implicitly presumes that the ratio between desirable and undesirable outputs fluctuates, a condition that does not hold true in practice (Chen & Golley, 2014). Consequently, many recent studies have shifted to using the non-radial and non-oriented Slack-Based Measure (SBM) model proposed by Tone (Tone, 2001). Building on this, Tone (2002) introduced the super-efficiency SBM model, integrating the strengths of the super-efficiency Data Envelopment Analysis (DEA) and the SBM approaches. Lately, the super-efficiency SBM has become one of the most widely used and robust tools for

assessing green productivity growth, as evidenced by various studies (Guo & Yuan, 2020; Liu et al., 2021; Wu & Zhang, 2020; Yu et al., 2019).

In this paper, GTFP is evaluated by applying the super-efficiency SBM model with undesired outputs proposed by (Tone, 2002). The advantages of this method are the following three points:

- (1) A cost that aligns with actual manufacturing procedures is involved when reducing undesirable output.
- (2) By incorporating slack variables into the objective function, super-efficiency SBM eliminates radial and directional biases associated with traditional DEA (Song et al., 2012).
- (3) Using the super-efficient SBM model, multiple effective Decision Making Units (DMUs) can be ranked efficiently (Guo & Yuan, 2020).

For more details about the SBM model, please refer to Appendix A of the supplementary document.

After calculation, Fig. 3 depicts the changes in GTFP levels over time and space among different countries. The GTFP levels in Africa exhibit a diverse trend, with some countries showing significant improvement in specific years, leading to an overall upward trend. Countries in Asia, including China and India, have demonstrated continuous GTFP growth in recent decades, which reflects a continuous improvement in green production efficiency. European countries had relatively high GTFP levels in the early 1990s, with overall stable changes; however, countries like Greece and Italy experienced varying degrees of fluctuation during specific periods. South and North America display different trends, with Brazil and Argentina showing some fluctuations in growth rates. At the same time, Canada exhibited noticeable fluctuations in specific years and an overall upward trend. From 1991 to 2022, most countries worldwide have seen improvements in GTFP, though the rate and extent of growth vary. Developing countries in Asia have shown remarkable performance in green production efficiency, whereas countries in Europe and the Americas have exhibited diverse development trends. Future research should focus on the impact of national policies and practices on GTFP to provide more targeted recommendations and improvement measures.

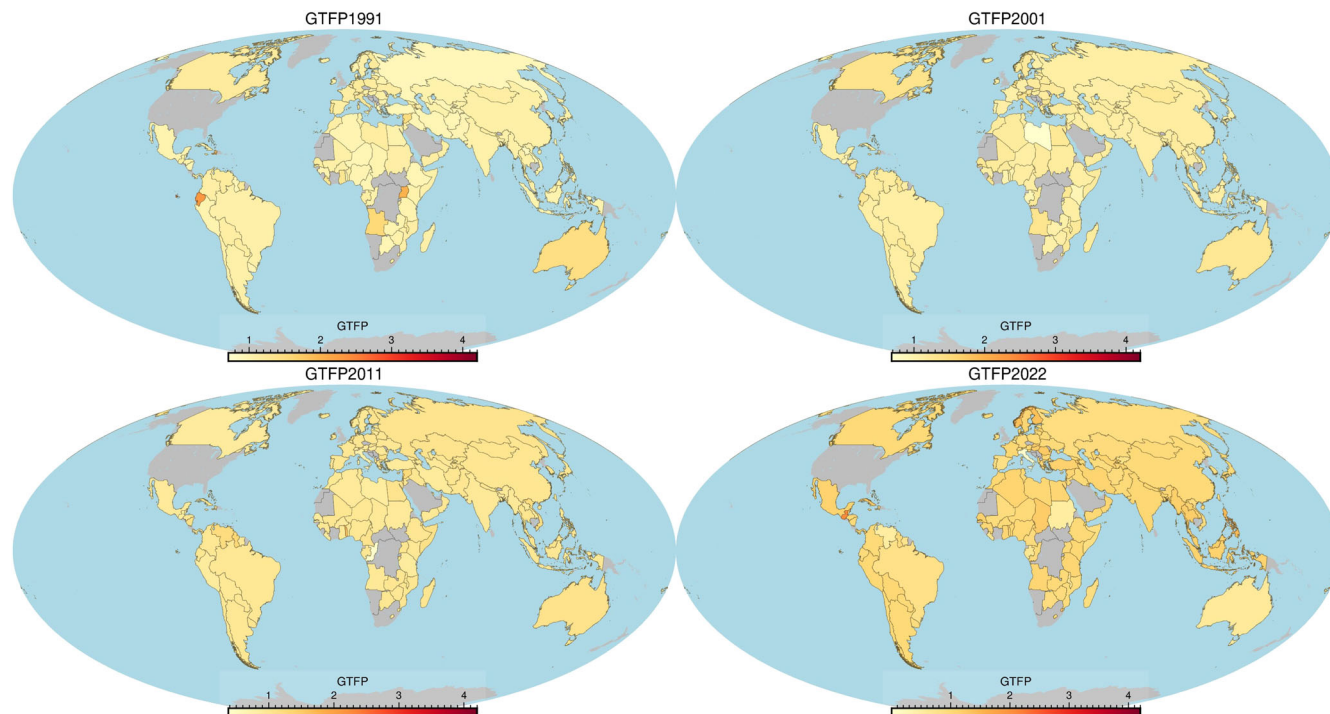


Fig. 3 Temporal and spatial evolution of green total factor productivity levels.

Indicators for building a CSFM. Based on Chinese President Xi Jinping's elaboration on the CSFM at the 70th United Nations General Assembly, the core principles of this concept include equal and mutually beneficial partnerships, a fair and shared security framework, an open, inclusive, and innovative development outlook, the promotion of multicultural exchanges to foster harmony and mutual learning, and a green and sustainable development model (Xi, 2015). Therefore, the quantitative index for the CSFM can be constructed based on these five aspects. Stott & Murphy (2020) explored the potential of cooperative partnerships in achieving sustainable development goals from a relational perspective, indicating that establishing equal and mutually beneficial partnerships is crucial for sustainable transformational cooperation. Banerjee et al. (2020) emphasized the importance of multi-stakeholder cooperative partnerships in promoting the attainment of sustainability objectives through case studies. Lee et al. (2020) used the 2019 Manila water crisis to demonstrate how fair and shared resources can enhance water security, thereby supporting sustainability. Rockström et al. (2023) discussed the importance of building a more equitable and inclusive recovery mechanism post-COVID-19, which is vital for driving sustainable economic growth and social resilience. By analyzing the significance of cooperative partnerships and the practical impacts of resource sharing, the studies illustrate how equitable transactions and optimized resource allocation can promote overall societal well-being and sustainability. Cai (2020) examined various aspects of sustainable development, including urban climate adaptation and inclusive environmental governance, highlighting the significance of fair opportunities and inclusion in reaching sustainable development objectives. Phonthanukitithaworn et al. (2023) analyzed how small and medium-sized companies can leverage open innovation and intellectual capital to drive sustainable development. Fatma & Haleem (2023) demonstrated the link between eco-innovation and sustainable development, showing how innovation can help reduce environmental risks and drive sustainable economic growth. These studies highlight how open innovation and intellectual capital can bring

sustainable company models to small and medium-sized companies and how eco-innovation can mitigate environmental risks and promote economic growth, supporting comprehensive sustainable development goals. Skrefsrud (2022) emphasized the role of education in promoting social sustainability through the application of multicultural education. Banban (2018) demonstrated how multicultural exchanges can enhance community harmony and sustainable development by studying the multi-ethnic cultures in Qinghai Province. These examples of multicultural education and the harmonious coexistence of multiple ethnicities illustrate how cultural diversity and inclusivity can enhance social sustainability through education and social practices. Wen & Dai (2021) proposed how advancements in green technology can enhance green GDP growth. OECD (2011) explained how green growth can ensure the sustainable use of natural resources, thereby supporting global sustainability. These examples of green technological advancements and green growth strategies show how environmental technologies and strategies can protect natural resources without compromising economic growth, supporting long-term sustainable development. Building a CSFM can promote the sustainable development of human society.

Table 1 lists the specific indicator system for the CSFM. Due to significant data gaps before 1991, after 2022, and for certain countries, the dataset used in this study is a panel dataset covering 184 countries and regions from 1991 to 2022. Some scholars have attempted to micro-analyze macro problems using data from around a decade, which can lead to substantial errors. Additionally, considering that this research seeks to investigate the pathways of human sustainable development, relying on data from only a few countries may lead to sample selection bias and geopolitical issues. Based on these considerations, this study integrates data from many countries over a long time series to comprehensively analyze and reveal one of the pathways to human sustainable development, the CSFM, thereby reducing errors and enhancing the generalizability and reliability of the research findings. The calculation method for the indicators

Table 1 Indicator System.

Category	Measurement	Attributes	Data Source
Equal and Mutual Benefit Partnership	Number of bilateral and multilateral trade agreements	+	World Trade Organization (WTO)
	Number of diplomatic visits and high-level meetings	+	United Nations (UN)
	Number of memberships in major international organizations	+	United Nations (UN)
	Average duration of memberships in major international organizations	+	United Nations (UN)
	Attendance rate at annual or special meetings of major international organizations	+	United Nations (UN)
	Number of speeches or proposals made at international organization meetings	+	United Nations (UN)
	Number of proposals adopted or passed	+	United Nations (UN)
	Number of high-level leadership positions held in major international organizations	+	United Nations (UN)
	Average tenure of high-level leadership positions	+	United Nations (UN)
	Number of major meetings chaired in international organizations	+	United Nations (UN)
	Number of major tasks and special projects led in international organizations	+	United Nations (UN)
Just and Shared Security Framework	Number of international military cooperation and joint exercises	+	United Nations Peacekeeping Database(UNPD)
	Number of personnel and financial contributions to international peacekeeping operations	+	United Nations Peacekeeping Database(UNPD)
	Number of signed international security agreements and treaties	+	United Nations Peacekeeping Database(UNPD)
	Number of reports submitted to the UN on the implementation of international security agreements and treaties	+	United Nations Peacekeeping Database(UNPD)
Open, Inclusive, and Innovative Development Prospects	Ratio of import and export volume to GDP	+	World Bank
	Foreign direct investment	+	United Nations Conference on Trade and Development (UNCTAD)
	Income inequality index (Gini coefficient)	-	World Bank, UNDP
	Ratio of R&D expenditure to GDP	+	Organisation for Economic Co-operation and Development (OECD)
	Ratio of high-tech product exports to total exports	+	World Trade Organization (WTO)
Multicultural Exchange Promotes Harmony and Mutual Learning	Number of World Heritage sites	+	United Nations Educational, Scientific and Cultural Organization (UNESCO)
	Number of cultural exchange programs and activities	+	United Nations Educational, Scientific and Cultural Organization (UNESCO)
	Number of cross-cultural education and academic exchange programs	+	United Nations Educational, Scientific and Cultural Organization (UNESCO)
	Number of international cultural exchange agreements	+	United Nations Educational, Scientific and Cultural Organization (UNESCO)
	Number of students studying abroad	+	United Nations Educational, Scientific and Cultural Organization (UNESCO)
	Proportion of population using basic drinking water services	+	United Nations Children's Fund (UNICEF), World Health Organization (WHO)
	Coverage rate of safely managed drinking water services and adequately managed sanitation facilities	+	United Nations Children's Fund (UNICEF), World Health Organization (WHO)
Green Sustainable Development Model	Number of national climate change policies and plans formulated and implemented	+	NDCs Database maintained by Maintained Nationally Determined Contributions (NDCs) Database
	Proportion of forest area to total land area	+	Food and Agriculture Organization (FAO) Global Forest Resources Assessment
	Proportion of degraded land area to total land area	+	United Nations Convention to Combat Desertification (UNCCD)
	Amount of nutrients and plastic waste discharged into the ocean annually	-	United Nations Environment Programme (UNEP)
	Proportion of marine protected area to national exclusive economic zone (EEZ) area	+	International Union for Conservation of Nature (IUCN)
	Carbon emissions	-	International Energy Agency (IEA), World Bank
	Number of clean energy patents	+	World Intellectual Property Organization (WIPO)
	Proportion of renewable energy in energy consumption	+	International Energy Agency (IEA), World Bank

utilizes the entropy method, a multi-indicator evaluation method used for comprehensive evaluation and weight estimation. It relies on the concept of entropy in information theory, determining the weight of each indicator in the comprehensive evaluation by calculating the entropy value of the indicators (Wu et al., 2022). The core concept of the entropy method is that the greater the variability of an indicator among the evaluated objects, the higher its information content and weight; conversely, the smaller the variability, the lower its information content and weight. For more technical details on the entropy method, please refer to Appendix B of the supplementary document.

The entropy method can objectively determine the weight of indicators, avoiding the influence of subjective human factors. It also considers the variability and information content of each indicator, helping to enhance the scientific validity of the evaluation results. The entropy method is widely used in multi-indicator comprehensive evaluations in economics, management, and environmental science (Olbrys, 2022; Wang et al., 2022; Zou et al., 2023). Figures 4 and 5 illustrate the changes in the CSFM index. By observing these figures, the trend can be divided into four periods: two periods of increase and two periods of decline. Analyzing these periods reveals the changes in the CSFM trend globally and across different continents at various times. This is significant for understanding the progress and challenges in building a global CSFM. Combining the indicator system with the actual conditions of different continents allows for a more differentiated explanation of these changes.

Considering the overall trend, the CSFM experienced significant fluctuations between 1991 and 2022. From 1991 to 1997, the CSFM declined from 0.59 to 0.57, possibly reflecting the political and economic instability following the end of the Cold War (UN Secretary-General & UN Secretariat, 1997). From 1997 to 2007, the CSFM increased from 0.57 to 0.59, a trend likely associated with accelerated globalization and strengthened international cooperation (United Nations, 2002). However, between 2007 and 2013, the CSFM declined again, from 0.59 to 0.58, which could be linked to the influence of the worldwide financial crisis (World Bank, 2012). From 2013 to 2022, the CSFM continuously increased from 0.58 to 0.62, reflecting the international community's efforts and achievements in addressing worldwide challenges, comprising progress in climate change mitigation, poverty reduction, and technological innovation (WTO, 2022).

From a regional perspective, Region I (1991–1997) shows varying performances across different continents. In Asia, the index first rose and then declined (as shown in Curve “a” in Fig. 4), reflecting the rapid economic growth post-Cold War, but it was additionally influenced by the Southeast Asian financial crisis. The increase in bilateral and multilateral trade agreements within the indicator system promoted economic development, but political instability impacted the overall CSFM (UN, ESCAP, 1996). The index rose and then slightly declined in Africa (as shown in Curve “b” in Fig. 4), showing short-term economic growth driven by international aid and resource development. However, persistent political turmoil and conflicts decreased the index (Collier & Gunning, 1999; Collier & Hoeffler, 1998). The number of international military cooperation and peacekeeping actions in the indicator system increased significantly (Doyle & Sambanis, 2000). Europe showed a stable and slightly rising overall trend (as shown in Curve “c” in Fig. 4), indicating the region's swift adaptation to the new international order post-Cold War, with high economic and political stability (Baker et al., 1997). Joining major international organizations and holding high-level leadership positions increased, enhancing its international status (Ginsberg, 1997). North America showed minor fluctuations but remained generally stable (as shown in Curve “d” in Fig. 4), demonstrating the region's economic resilience and

political stability during the initial phase of globalization. The high proportion of R&D expenditure to GDP maintained its innovation drive (OECD, 1998). The index showed a downward trend in South America (as shown in Curve “e” in Fig. 4), reflecting the region's political and economic challenges, including economic reforms and debt crises. The reduction in international cultural exchange projects impacted social harmony (Smith, 1994). Oceania's index continuously declined (as shown in Curve “f” in Fig. 4), showing the region's developmental difficulties in the early stages of globalization, particularly in dealing with environmental changes and economic dependency (World Bank, 1991).

During Region II (1997–2007), Asia's index fluctuated upward (as shown in Curve “g” in Fig. 4), reflecting the region's volatility amid rapid economic development. The proportion of high-tech products in total exports increased, but the region also experienced the 1997 Asian financial crisis (Heo & Hahm, 2007). Africa's index declined and then rose (as shown in Curve “h” in Fig. 4), showing significant fluctuations and uneven economic development. The number of participants and financial input in international peacekeeping operations increased, but infrastructure and public services still needed to be increased (Harmse, 2007). Europe's index mainly remained stable with slight fluctuations (as shown in Curve “i” in Fig. 4), indicating the region's stability during this period, particularly in economic cooperation and active participation in international organizations (Savage & Verdun, 2007). North America showed minor fluctuations and remained generally stable (as shown in Curve “j” in Fig. 4), demonstrating the region's leading role in promoting international trade and technological innovation (Schwab & Porter, 2006). In South America, the index first rose and then declined (as shown in Curve “k” in Fig. 4), showing significant variability and reflecting the region's economic instability, particularly in facing external debt issues and political turmoil (UN, ECLAC, 2007). Oceania's index showed significant fluctuations but an overall upward trend (as shown in Curve “l” in Fig. 4), reflecting the region's gradual adaptation and development in globalization, especially in cultural exchange and environmental protection (Dreher, 2006; Young, 2002).

Region III (2007–2013) highlights the impact of the global financial crisis on various continents. During this period, Asia's index declined (as shown in Curve “a” in Fig. 5), reflecting the global financial crisis's effects despite increased regional economic cooperation (Nasreen & Anwar, 2014). Africa's index first rose and then declined (as shown in Curve “b” in Fig. 5), showing the region's volatility during the crisis, particularly the vulnerability of its resource-dependent economic structure (Khan, 2014). Europe's index remained relatively stable (as shown in Curve “c” in Fig. 5), indicating relative stability during the crisis, thanks to its robust economic and financial systems (Cohen, 2012). North America's index showed a downward trend (as shown in Curve “d” in Fig. 5), reflecting the profound impact of the financial crisis, especially when the real estate market and financial institutions were severely hit (Neaime, 2012). South America's index declined (as shown in Curve “e” in Fig. 5), reflecting the region's difficulties during the crisis, mainly due to capital outflows and commodity price fluctuations (Kumar, 2013). Oceania's index showed a downward trend (as shown in Curve “f” in Fig. 5), indicating the financial crisis's impact on the region, especially regarding foreign trade and tourism (Vera et al., 2014).

Region IV (2013–2022) showcases the recovery and development of various continents in the post-financial crisis period. During this period, Asia's index continuously rose (as shown in Curve “g” in Fig. 5), reflecting the region's recovery and development, mainly through the BRI that promoted regional cooperation and

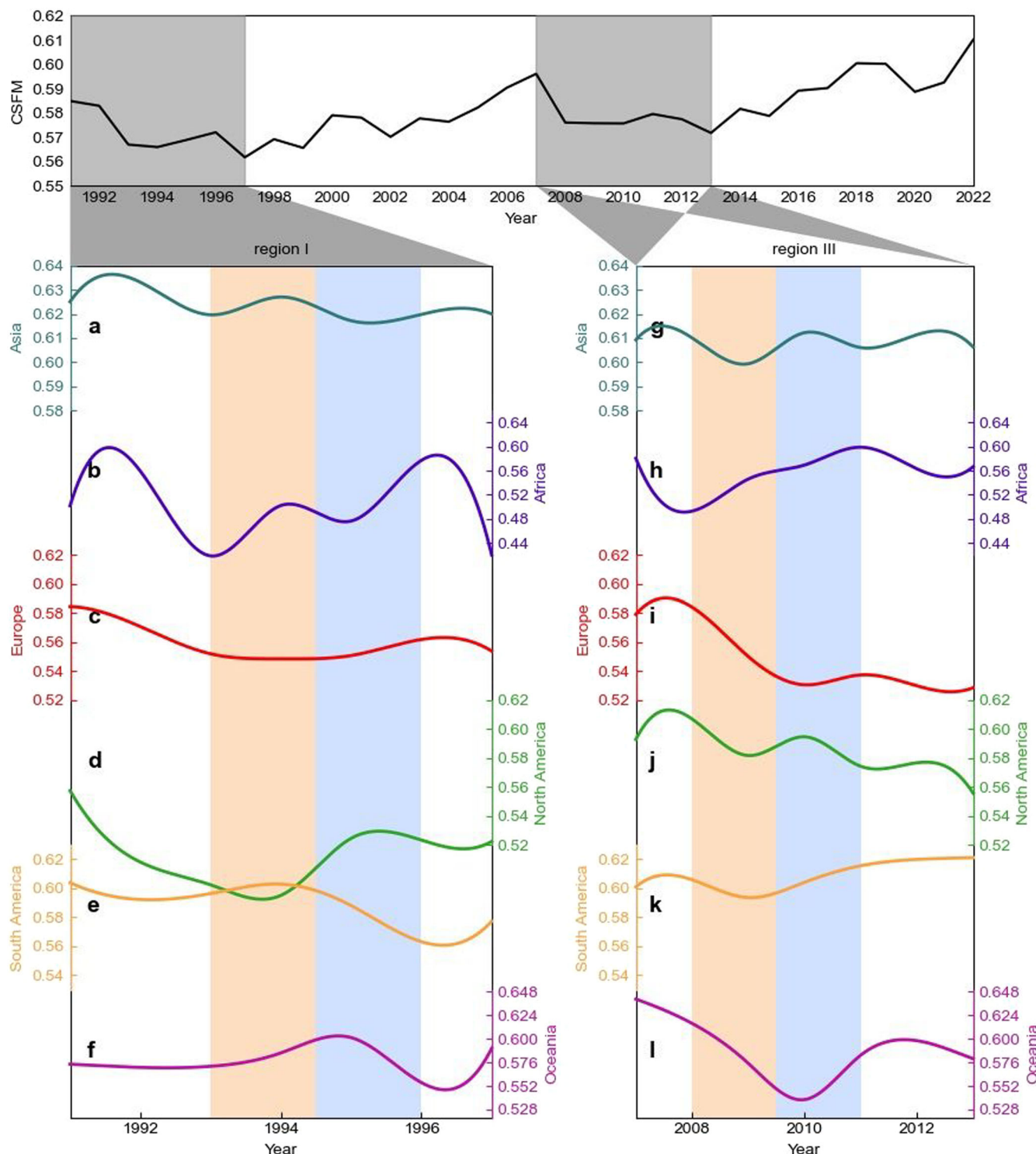


Fig. 4 Visualization of the CSFM Index 1. **a** represents the index for Asia during Region I (1991–1997). **b** shows Africa's index during Region I. **c** displays Europe's index. **d** reflects North America's index. **e** represents South America's index. **f** illustrates Oceania's index. For Region II (1997–2007): **g** shows Asia's index. **h** depicts Africa's index. **i** represents Europe's index. **j** shows North America's index. **k** illustrates South America's index. **l** shows Oceania's index.

infrastructure construction (Asian Development Bank, 2020; Jian, 2018; Zhang, 2021). Africa's index showed upward fluctuations with an overall positive trend (as shown in Curve "h" in Fig. 5), indicating progress in international aid and cooperation, especially with increased investments in infrastructure and education (UN Secretary-General, 2021). Europe's index displayed a stable upward trend (as shown in Curve "i" in Fig. 5), reflecting the region's

recovery and development, particularly in green sustainable development and social inclusion policies (APEC, 2015). North America's index fluctuated but generally rose (as shown in Curve "j" in Fig. 5), showing the region's recovery efforts, particularly in technological innovation and economic diversification (OECD, 2023). South America's index fluctuated significantly but generally showed an upward trend (as shown in Curve "k" in Fig. 5),

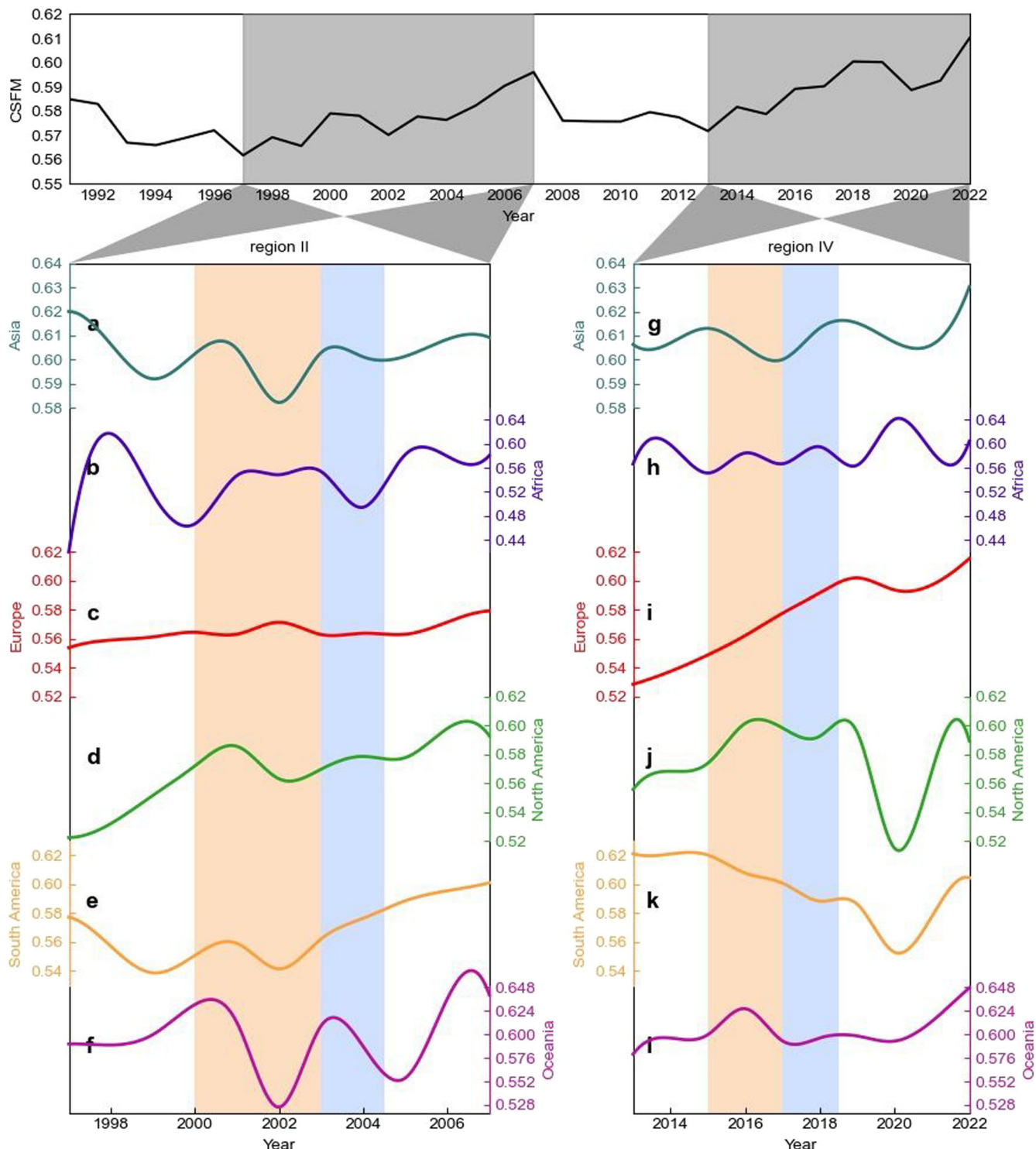


Fig. 5 Visualization of the CSFM Index 2. **a** represents the index for Asia during Region I (1991–1997). **b** shows Africa's index during Region I. **c** displays Europe's index. **d** reflects North America's index. **e** represents South America's index. **f** illustrates Oceania's index. For Region II (1997–2007): **g** shows Asia's index. **h** depicts Africa's index. **i** represents Europe's index. **j** shows North America's index. **k** illustrates South America's index. **l** shows Oceania's index.

reflecting the region's developmental efforts post-crisis, especially in social reforms and economic recovery policies (World Economic Forum, 2024). Oceania's index also fluctuated significantly but generally trended upward (as shown in Curve “l” in Fig. 5), indicating progress in the region's recovery, particularly in environmental protection and cultural exchange (UN. ESCAP et al., 2019).

The above analysis aligns with existing literature on globalization, international cooperation, and economic development disparities. For instance, the literature mentions that the increase in globalization and international trade positively impacts the construction of CSFM, while global events such as financial crises can have adverse effects. Additionally, the development disparities among regions are reflected in these

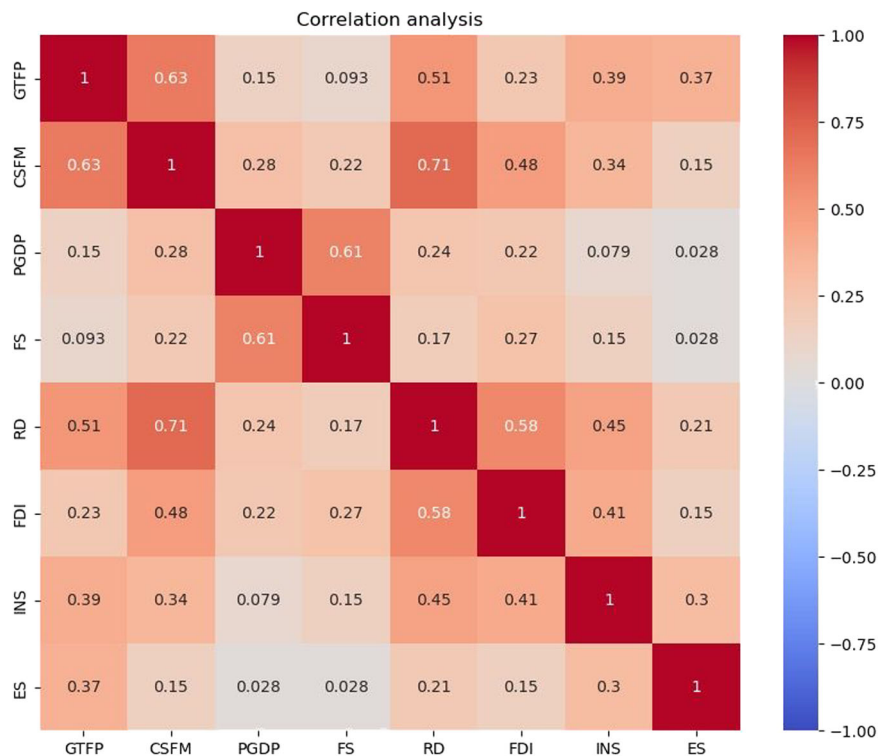


Fig. 6 Pearson Correlation Coefficient Matrix.

charts, showing different progress and challenges faced by various regions in globalization and international cooperation. Through analyzing these two figures, a deeper insight into the shifts in the global CSFM index and its driving factors can be achieved, providing a basis for further policy formulation and international cooperation.

Econometric model. This study employs the following econometric model to analyze the influence of the CSFM on the sustainable development of human society.

$$GTFP_{it} = \alpha_0 + \alpha_1 CSFM_{it} + X_{control} \alpha_2 + C_i + \lambda_t + trend_j + \epsilon_{it}$$

(1)

Here, *i* and *t* represent countries and years, respectively. GTFP indicates sustainable development, while CSFM denotes the CSFM. *X_{control}* represents control variables, which are selected based on the studies by Cheng et al. (2018); Wu et al. (2020, ?); Yuan et al. (2020). These control variables include per capita GDP (PGDP) and fiscal support (FS) to measure scale effects, R&D investment and foreign direct investment (FDI) to measure technological effects, and industrial structure (INS) and energy consumption structure (ES) to measure structural effects. *λ_t* represents time fixed effects (TFE), and *C_i* represents country-level fixed effects that remain constant over time. Incorporating country- and time-fixed effects into the econometric model aims to enhance the accuracy and reliability of the analysis (Baltagi, 2008). Country-level fixed effects control unobservable characteristics among countries, such as culture, institutions, and geographic location, thereby reducing bias caused by these differences (Bell & Jones, 2015). Time-fixed effects to control for time effects that all countries experience during the study period, such as global economic crises and changes in international policies, ensuring that the impact of these time variations is not overlooked (Callaway & Sant’Anna, 2021). By introducing these two fixed effects, omitted variable bias can be reduced, specific impacts for each country and period variations can be

Table 2 Baseline Regression, Robustness Tests1.

	OLS	PLS	MLE	Ridge	Lasso
CSFM	0.5105*** [8.1196]	0.4263*** [3.8584]	0.5212*** [4.3952]	0.5243*** [3.6201]	0.4928*** [3.9164]
_cons	0.6892*** [3.8323]	0.6388** [2.1780]	0.6886* [2.3837]	0.6224*** [3.3409]	0.6215*** [3.3075]
control	YES	YES	YES	YES	YES
FE	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES

, and * indicate p-values less than 5%, and 1%, respectively, equivalent to confidence levels of 95%, and 99%. From a statistical perspective, a confidence level above 90% is necessary to assert the correctness of the results. The values in parentheses represent t-values, which are used to calculate p-values and confidence levels. The values above the parentheses are regression coefficients. A positive regression coefficient indicates that the CSFM promotes the sustainable development of human society. _cons represents the intercept term of the model. Control indicates whether control variables are included in the model. FE denotes whether country-level fixed effects are included. Year indicates whether time-fixed effects are included. The statistical results were obtained using Stata.

captured, and the model’s fit and explanatory power can be improved, resulting in more reliable research findings (de Chaisemartin and D’Haultfœuille, 2024).

Results and discussion

To examine the impact of the CSFM concept on the “sustainable development of human society,” this section conducts a regression analysis based on the indicators constructed in Section 3. This section covers baseline regression, robustness tests, endogeneity, and heterogeneity analysis. Baseline regression serves as the study’s starting point, preliminarily exploring the impact of independent variables on dependent variables. Robustness tests verify the reliability and stability of the baseline regression results. The stability of the results can be assessed by employing different regression methods and altering model specifications. Endogeneity analysis addresses endogeneity issues in the regression

Table 3 Robustness Tests2.

	OLS	PLS	MLE	Ridge	Lasso
replace GTFP with GTFP1	0.4522*** [4.6972]	0.4605*** [3.5662]	0.4485** [2.5476]	0.4611** [2.1764]	0.4778** [2.3182]
replace CSFM with CSFM1	0.5491*** [4.6727]	0.5261** [2.4641]	0.5617** [2.4976]	0.5223** [2.0997]	0.5011** [2.1682]
control	YES	YES	YES	YES	YES
FE	YES	YES	YES	YES	YES
Year	YES	YES	YES	YES	YES

, and * indicate p-values less than 5%, and 1%, respectively, equivalent to confidence levels of 95%, and 99%. The relevant explanations are as in Table 3. The regression coefficients in the table represent the regression results of CSFM on the sustainable development of human society. The two rows report the results of replacing GTFP with GTFP1 and CSFM with CSFM1, respectively. The statistical results were obtained using Stata.

model, which may bias the estimation results; thus, endogeneity treatment is necessary to ensure the validity of the results. Heterogeneity analysis explores the discrepancies in the influence of independent variables on dependent variables across various subsamples, examining the generalizability of the research findings.

Baseline regression and robustness tests. Figure 6 shows the Pearson correlation matrix between CSFM and the sustainable development of human society. The correlation coefficient between CSFM and sustainable development of human society is 0.63, indicating a significant positive correlation. This finding preliminarily verifies that building a CSFM can promote the sustainable development of human society. Table 2 presents the regression results of the CSFM and sustainable development of human society from 1991 to 2022 across models (1)–(5). Due to the different assumptions and applicable conditions of various estimation methods, using multiple methods can verify the robustness of the results. The results are considered robust and reliable if conclusions from different methods are consistent. Model (1) employs Ordinary Least Squares (OLS) estimation, model (2) uses Partial Least Squares (PLS) estimation, model (3) applies Maximum Likelihood Estimation (MLE), model (4) uses Ridge regression, and model (5) employs Lasso regression. Using OLS as the baseline regression method, followed by robustness checks with PLS, MLE, Ridge Regression, and Lasso Regression, ensures the model's robustness and reliability. OLS provides a benchmark result, and comparing this with the results from PLS, MLE, Ridge, and Lasso helps verify the consistency of the findings. PLS improves the predictive accuracy of high-dimensional data through dimension reduction and projection. MLE offers more flexibility and adaptability in parameter estimation. Ridge and Lasso enhance model robustness and prevent overfitting by introducing regularization, effectively addressing multicollinearity issues. The regression outcomes in Table 2 consistently indicate that the CSFM promotes the sustainable development of human society. Moreover, the outcomes are statistically important at the 1% level (with a confidence level exceeding 99%).

However, more than merely substituting regression methods for robustness checks are required. The concretization of the abstract concepts of CSFM and sustainable development of human society for analysis inherently involves subjectivity. Therefore, this study employs different measurement methods to recalculate the abovementioned indicators. If the conclusions remain unchanged, it demonstrates the robustness of this study's findings. Specifically, following the study by Gewers et al. (2021), another commonly used objective weighting method—Principal Component Analysis (PCA)—is used to recalculate the CSFM indicator, denoted as $CSFM_1$. The proxy variable for the sustainable development of human society is GTFP. Following

Table 4 Endogeneity Analysis.

	SYS-GMM	DIF-GMM
L.GTFP	0.8536*** [3.4896]	0.9237*** [3.6121]
CSFM	0.4953*** [5.3803]	0.4866*** [5.5900]
control	YES	YES
FE	YES	YES
Year	YES	YES
AR(1)	P-value = 0.021	P-value = 0.015
AR(2)	P-value = 0.857	P-value = 0.624
Hansen test	P-value = 0.742	P-value = 0.953
Sargen test	P-value = 0.715	P-value = 0.816

*** indicates p-values less than 1%, which are equivalent to confidence levels of 99%. L.GTFP represents the lagged term of GTFP. The statistical results were obtained using Stata.

the study by Chen et al. (2023), the Malmquist index method recalculates this indicator, denoted as $GTFP_1$. The regression outcomes, as shown in Table 3, indicate that even after substituting the proxy variables, the influence of CSFM on the sustainable development of human society remains positive, with all results having a confidence level exceeding 95%. In summary, the research conclusions are robust.

Endogeneity analysis. Following the approach of Anderson & Hsiao (1982); Arellano & Bond (1991), this study selects appropriate instrumental variables and employs both system GMM and different GMM methods. These methods estimate the model based on the first-order lag of CSFM to test for endogeneity issues, ensuring the validity of the association between CSFM and the sustainable development of human society discussed earlier. Before applying the system GMM or difference GMM models for empirical analysis, a series of important tests must be conducted to ensure the model's validity and reliability. First, the Sargan and Hansen tests are used to evaluate the validity of the instrumental variables. The Sargan test (also known as the over-identification restriction test) primarily tests the independence of all instrumental variables from the error term, ensuring the validity of the instruments. A large p-value from the Sargan test indicates valid instruments; conversely, a small p-value (less than 0.1) suggests that the instruments may be invalid and correlated with the error term. The Hansen test (also known as Hansen's J test) is more robust in handling heteroskedasticity and serial correlation. The Hansen test similarly assesses the instruments' validity and the model's correct specification. A large p-value shows that the instruments are valid and the model is correctly specified, whereas a small p-value (less than 0.1) implies invalid

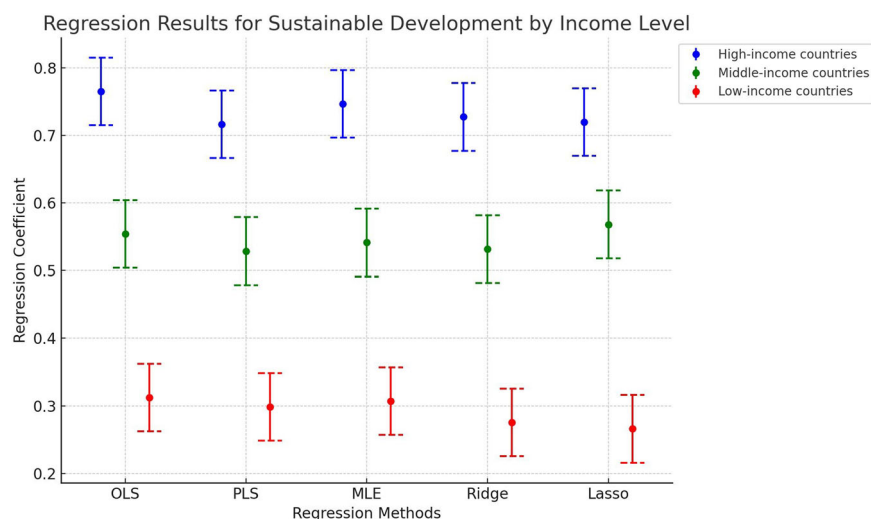


Fig. 7 Heterogeneity Analysis results.

instruments or incorrect model specification. Table 4 shows that the p-values for both the Sargan and the Hansen tests are significantly greater than 0.1, indicating that the instrumental variables chosen in this study are valid. Second, the first-order autoregression test (AR(1) test) and the second-order AR(2) test are used to detect the AR of the error term. In systems, GMM, and different GMM models, the independence of the error term is an important assumption. The AR(1) test detects first-order AR, i.e., the correlation between the current error term and the error term of the previous period. A large p-value suggests no significant first-order AR. The AR(2) test detects second-order AR, i.e., the correlation between the current error term and the error term of the period before the last. According to research, the system GMM model should satisfy AR(1) (p-value less than 0.1) but not AR(2) (p-value more significant than 0.1). Table 4 presents the regression outcomes. In this study, the instrumental variables pass both the Sargan and Hansen tests, and the models meet the AR(1) requirement while exhibiting no AR(2). By estimating the overall sample using the system GMM and different GMM models, it is found that CSFM can enhance the level of sustainable development of human society, indicating that the regression models discussed earlier are valid.

Heterogeneity analysis. Through regression analysis of data from 184 countries and regions worldwide, this paper finds that the promotion impact of the CSFM on the sustainable development of human society is significant and robust. To further verify the universality of this conclusion, the sample countries are categorized into high-income, middle-income, and low-income groups for heterogeneity analysis. As illustrated in Fig. 7, the findings from the heterogeneity analysis indicate that the CSFM has the most significant positive impact on sustainable development in high-income countries. This result can be attributed to high-income countries having well-developed infrastructure, high levels of technology, and sound policy systems, enabling them to implement and benefit from the CSFM more effectively. Additionally, these countries have advantages in international cooperation and technological innovation, allowing them to fully leverage the opportunities brought by the CSFM to promote sustainable development further. For example, Chan et al. (2022) noted that high-income countries' infrastructure and technological advantages facilitate the implementation of international cooperation projects, thereby advancing sustainable development goals. The CSFM exhibits a positive promoting effect for middle-

income countries, though to a lesser extent than in high-income countries. These countries rapidly develop and can gain significant technology transfers, investments, and international cooperation opportunities from the CSFM (Bhattacharya et al., 2015). However, the relative insufficiency of infrastructure and technological levels and the need for policy adjustments may limit the full potential of the CSFM. Nevertheless, middle-income countries can significantly enhance their sustainable development levels through appropriate policy adjustments and international support. Saha et al. (2022) emphasized in their research that middle-income countries need to optimize their policy environment to attract international investment and technological support better, thereby promoting sustainable development. In low-income countries, the effect of the CSFM is relatively small or insignificant. This result is mainly due to weak infrastructure, lack of funds and technology, and incomplete policy and governance structures in these countries (Zhan & Santos-Paulino, 2021). Low-income countries need help to fully utilize the opportunities the CSFM brings to improve sustainable development significantly. Therefore, these countries require more international aid and technological support to improve infrastructure, enhance technological levels, and strengthen policy and governance systems, thereby better building the CSFM and promoting sustainable development.

In summary, the heterogeneity analysis indicates that the effect of the CSFM on sustainable development varies significantly across countries with diverse income levels. High-income countries benefit the most, followed by middle-income countries, while the effect is the smallest in low-income countries. This suggests that in building the CSFM, it is essential to fully consider the economic development levels and specific conditions of different countries, formulating differentiated policies and measures to ensure that the CSFM can maximize its promoting effect globally, thereby advancing the sustainable development of human society.

Conclusions

The five quantitative indicators of the CSFM significantly promote sustainable development of human society. These indicators encompass the core concepts of the CSFM: equitable and mutually beneficial partnerships, a just and shared security framework, an open, inclusive, and innovative development perspective, multicultural exchanges promoting harmony and mutual learning, and a green and sustainable model.

Consequently, it can be concluded that the CSFM fosters the sustainable development of human society and is a viable approach to achieving this goal. Building the CSFM can effectively promote the sustainable development of human society and lay a solid foundation for global well-being.

Drawing from the research above, this paper presents the following policy recommendations:

Enhancing International cooperation and coordination, promoting multicultural exchange, and fostering value recognition of the CSFM. Countries should strengthen international cooperation and coordination by signing bilateral and multilateral cooperation agreements to enhance economic, environmental, security, and cultural cooperation, promoting resource sharing and mutual benefit. Nations should bolster cultural exchange programs, encouraging cultural interaction and mutual learning among countries and regions with diverse cultural backgrounds to foster cultural diversity and inclusivity. Hosting international cultural festivals, art exhibitions, and academic exchanges can promote mutual understanding and appreciation of different cultures. Through study abroad programs, faculty and student exchanges, and collaborative research, international cooperation in education can be enhanced, cultivating talents with a global perspective. International cooperation should also focus on protecting and preserving cultural heritage and respecting and promoting cultural diversity. By reducing cultural conflicts through various measures, human society can enhance mutual understanding and value recognition, thus building a harmonious international community.

Strengthening fair and shared security frameworks, innovating development, and building institutional frameworks for the CSFM. Economic globalization, digitalization, and other factors have led to conflicts between nations, including economic competition, trade disputes, desires to control strategic lands or seas, and religious or ideological conflicts. The law of the jungle makes it challenging to achieve a state of shared benefits in human society. Countries should establish and improve international security cooperation mechanisms, promoting a just and reasonable international security order to jointly address terrorism, cybersecurity threats, natural disasters, and other non-traditional security challenges. Increased investment in scientific research and technological innovation should promote an open and inclusive spirit of cooperation to drive economic growth and social progress. All countries and regions need to formulate relevant policies to resolve these disputes. As humanity pursues commonly recognized values and interests, it is necessary to build mutually accepted institutions to prevent scenarios where political, administrative, economic, and technological power disparities lead to the intense bullying of the weak.

Promoting green sustainable development, improving infrastructure, and sharing the benefits of the CSFM. Joint development is the foundation of sustainable development, aligning with all people's long-term and fundamental interests globally. Different countries, transnational organizations, and ethnic groups have different interest entities. These entities can maximize their interests and share benefits within globally integrated social relations by engaging in mutual interactions, adhering to jointly established institutions, and pursuing common values. Governments should vigorously promote the formulation and implementation of green economic policies, encouraging green technology innovation and the development of green industries to reduce environmental pollution and improve resource utilization efficiency. According to heterogeneity analysis, differences in infrastructure levels may explain the varied impacts of the CSFM

on the sustainable development of different countries. High-income countries with well-developed Infrastructure benefit more. Middle-income countries need to accelerate modernization to enhance economic potential. In contrast, low-income countries should address infrastructure deficiencies by improving transportation, electricity, and water supply systems through international aid and cooperation to better integrate into the global economy and achieve sustainable development. Building the CSFM can not only promote policy convergence among countries, thus achieving trade and labor liberalization, but also facilitate capital exchange to address capital surplus issues, share the benefits of the community, and promote the sustainable development of human society.

Limitations and future research. This study has the following limitations. First, the subjectivity of the quantitative indicators is an important issue. Although this study attempts to quantify the five core principles of the CSFM and uses the super-efficiency SBM model from operations research to calculate the proxy variable for sustainable development, this quantification process may involve some subjectivity. Different scholars might have varying interpretations and quantification methods for these core principles, affecting the universality and consistency of the research results. Second, the availability and completeness of data are also limiting factors. While this study's dataset covers the period from 1991 to 2022 and includes 184 countries and regions, data for some countries may need to be completed, impacting the models' stability and the results' reliability. The lack of complete data limits the comprehensiveness and accuracy of the analysis. Lastly, the limitations of the mechanism analysis must be considered. This study primarily uses quantitative indicators and regression models to analyze the influence of the CSFM on sustainable development, but the specific mechanisms still need to be explored. For instance, implementing the core principles of the CSFM through concrete policies and measures requires further investigation and verification. Based on these limitations, future research can improve in the following areas. First, efforts should be made to optimize the quantification models and methods to enhance their accuracy and applicability. Exploring new quantitative indicators and methods can provide a more comprehensive assessment of the influence of the CSFM on the sustainable development of human society, particularly by reducing biases from subjectivity through more rigorous statistical methods and multidisciplinary theoretical support. Second, the temporal and spatial coverage of the data should be expanded further to verify the stability and universality of the research findings. It is essential to include more data from developing countries and different economies to explore the applicability of the CSFM under various levels of economic development and political systems. Additionally, efforts should be made to improve data completeness and accuracy, minimizing the influence of missing data on the analysis outcomes. Lastly, a more in-depth analysis of specific mechanisms is needed to explore how the core principles of the CSFM can be realized through concrete policies and measures. Combining case studies and empirical analysis can identify successful experiences and potential issues, providing policy recommendations for other countries. Integrating theories and methods from economics, sociology, political science, and environmental science will enable more comprehensive and systematic research. These future research directions will reveal more thoroughly and deeply the impact of the CSFM on sustainable development, providing more scientific and reliable evidence for global governance and policy-making.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

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